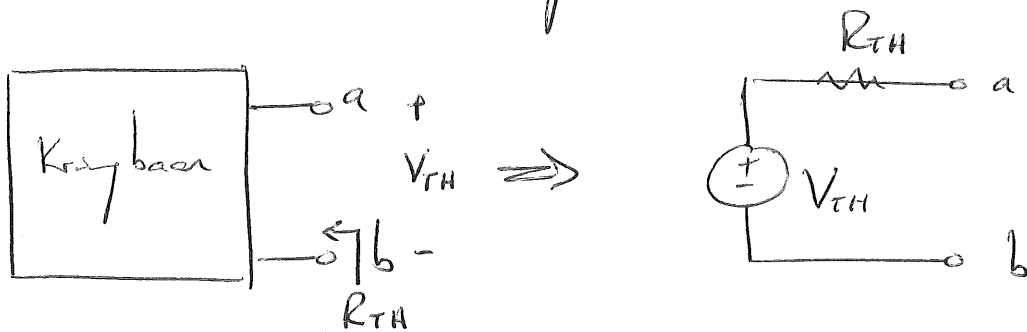


Thevenin's Theorem

→ Thevenin'se Theorie:

- * Vervang enige 2 terminaal kringbaar met
 " spanningsbron in series met " resistor
 zodat die twee kringbare equivalent
 is met betrekking tot die 2 terminale.



→ V_{TH}

- * Thevenin spanning is die spanning
 tussen terminale a en b.
 (Ope-baan spanning!)

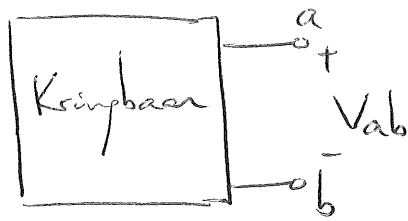
→ R_{TH}

- * Thevenin weerstand is die
 ekwivalente weerstand tussen die terminale
 indien alle onafhanklike bronne
 afgeskakel is.

* Hoe om V_{TH} en R_{TH} te bereken:

(2)

⇒ V_{TH}



* Los voor V_{ab} op dew:

* KVL; KCL; Σ -Wet

* Lusanalise; Nodanalyse

* Superpositie; Brontrand.

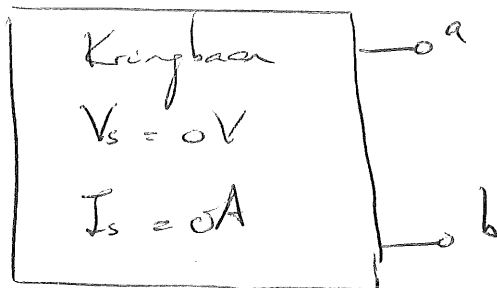
* $V_{oc} \Rightarrow$ 'open-circuit' spanning.

$$* V_{TH} = V_{ab} = V_{oc}$$

* V_{TH} is spanning tussen terminale a en b.

$$(V_{TH} = V_{ab} = V_a - V_b.)$$

⇒ R_{TH}



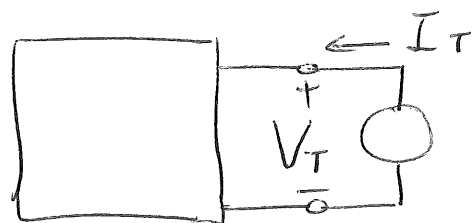
* Stel alle onafh bronne af.

* Indien daar geen afhanklike bronne is nie:

$$R_{TH} = R_{ab}.$$

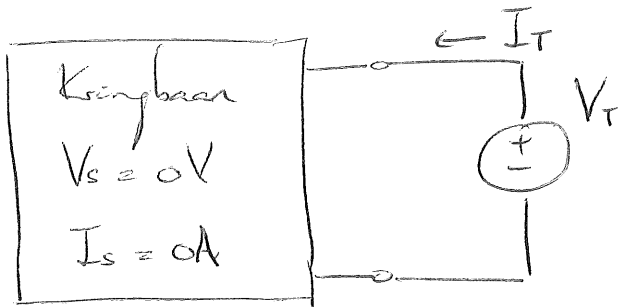
* Indien daar afh bronne is:

$$R_{TH} = \frac{V_{TOETS}}{I_{TOETS}} = \frac{V_T}{I_T}$$



Toetsbron!

⇒ Toetsbronne.



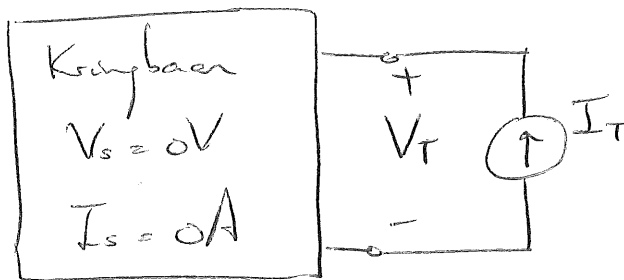
* Kies $V_T (= 1V)$

* Bepaal I_T .

$$R_{TH} = \frac{V_T}{I_T}$$

$$(\quad = \quad 1/I_T)$$

OF



* Kies $I_T (= 1A)$

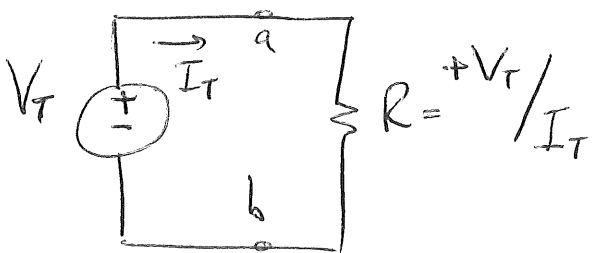
* Bepaal V_T

$$R_{TH} = \frac{V_T}{I_T}$$

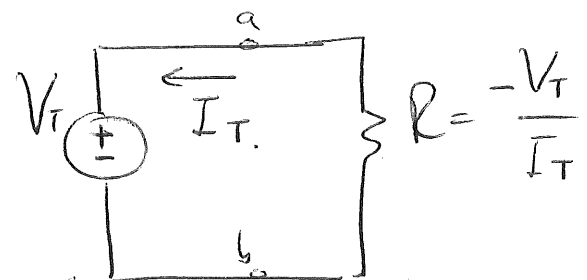
$$(\quad = \quad V_T/1)$$

* Kies spanningsbron of stroombron
afhangerende van voorkeurs.

* Let op: Richting van I_T .
Bv:



↳ Vloei weg van V_T .

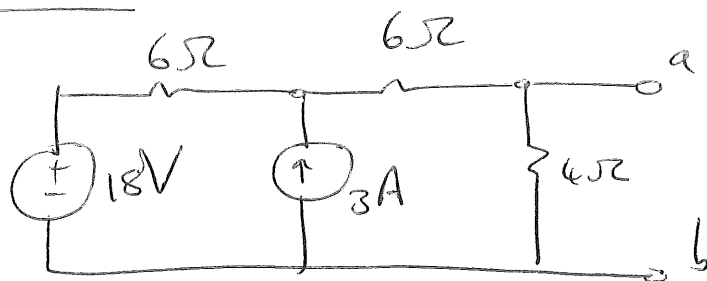


↳ Vloei na V_T .

Voorbeeld

④

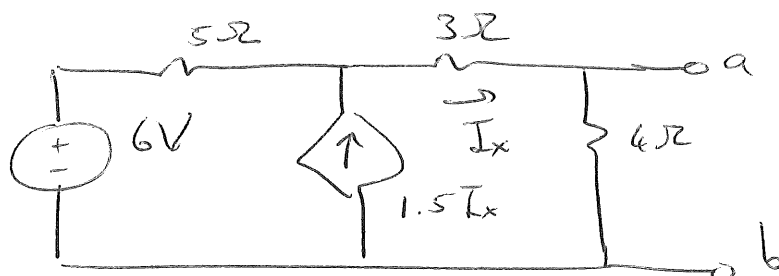
①



$$V_{TH} = ?$$

$$R_{TH} = ?$$

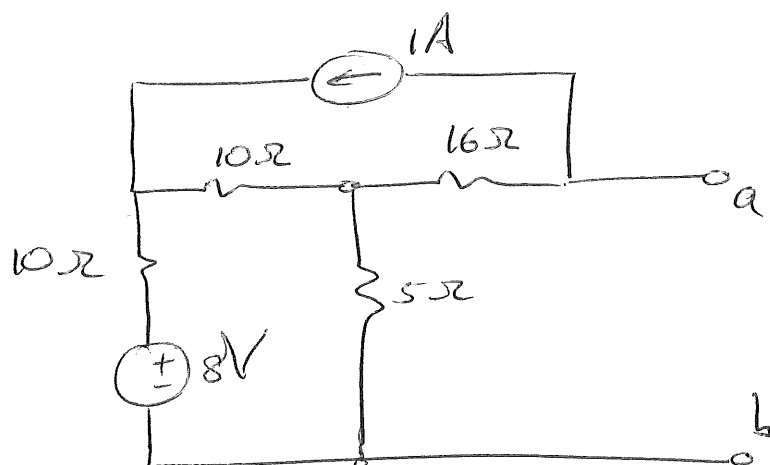
②



$$V_{TH} = ?$$

$$R_{TH} = ?$$

③



$$V_{TH} = ?$$

$$R_{TH} = ?$$